

MACHINE-CHECKED TRIAD IDENTITIES FOR THE FOURIER–GALERKIN TRUNCATION OF THE 3-D NAVIER–STOKES EQUATIONS, WITH THE VANISHING LOCUS OF THE HELICAL INTERACTION COEFFICIENT IN CLOSED FORM

AUTHOR LIST AND AFFILIATIONS TO BE SUPPLIED

ABSTRACT. We give a formally verified account of the quadratic nonlinearity of the incompressible 3-D Navier–Stokes equations restricted to the Fourier–Galerkin truncation $\Lambda_M = \{k \in \mathbb{Z}^3 : 0 < |k| \leq M\}$. (i) A single *weighted triad identity* yields exact energy conservation (constant weight) and the exact enstrophy production (weight $|k|^2$) as two instances of one computation, isolating the production mechanism in the factor $|r|^2 - |q|^2$, the wavenumber disparity of a triad’s two swapped members. (ii) For Waleffe’s helical decomposition we give the triad interaction coefficient in a square-root-free integer frame, prove that the resonance condition $s_k|k| + s_p|p| + s_q|q| = 0$ holds for some signs if and only if the triad is collinear (a fact of Euclidean geometry, independent of the lattice), and prove the “if and only if”: the coefficient vanishes exactly on the collinear triads and on the balance locus $s_p|p| = s_q|q|$, and nowhere else. (iii) On any periodic lattice without 2-torsion the resonant-triad incidence structure has an exactly solvable spectrum with normalised gap $\rightarrow 1$; on the spherical truncation the gap appears to converge to $5/6$, which we state as an open conjecture with six exact data points. Every theorem is machine-checked in Lean 4 against Mathlib with axiom footprint `{propext, Classical.choice, Quot.sound}` and no `sorry`; every closed form is independently cross-checked in exact integer arithmetic. The identities and the coefficient are classical (Kraichnan, Waleffe); what is new is the unified, machine-checked form, the explicit converse, and the exact spectrum. All results are at fixed M ; whether the production admits any bound uniform in M is not addressed.

1. INTRODUCTION

The Fourier–Galerkin truncation of the incompressible 3-D Navier–Stokes equations retains the Fourier coefficients u_k , $k \in \mathbb{Z}^3$, with $0 < |k| \leq M$ and discards the rest. The result is a finite-dimensional system whose global well-posedness is elementary, and whose quadratic nonlinearity keeps, exactly, the triad structure $p + q = k$ that carries the energy cascade and vortex stretching in the full equations. Detailed conservation over triads goes back to Kraichnan [1]; the helical decomposition, in which each mode splits into two chiralities and each triad interaction reduces to one complex coefficient, is due to Waleffe [2]. Nothing in this paper changes those facts. What it adds is a *formally verified* version of them, organised so that the structure is visible in the proofs rather than in the commentary, together with two statements we have not seen made explicit: an “if and only if” for the vanishing of the helical coefficient (Theorem 5.1), and the exact spectrum of the resonant-triad incidence structure on a periodic lattice (Theorem 6.1), whose spherical analogue we leave as a sharply posed conjecture.

The organising observation of Section 3 is that energy conservation and enstrophy production are the constant-weight and $|k|^2$ -weight instances of one identity on triads (Theorem 3.1): everything common to a triad’s two orderings cancels, and what remains is the difference of the weight at the two swapped members. For the energy weight the difference is zero, which is the conservation law; for the enstrophy weight it is the wavenumber disparity $|r|^2 - |q|^2$, which is the production mechanism, exactly.

Sections 4–5 treat the helical coefficient. Working in a frame built from the integer vector $p \times q$, and with helical vectors left unnormalised, the coefficient becomes a polynomial identity in integer variables and three formal magnitudes, so that it can be checked in exact integers over thousands of lattice triads — which caught a sign error in our first derivation that a formal proof would have certified just as happily (Remark 4.3). Section 6 gives the spectral result and the conjecture; Section 7 collects open questions.

Verification. Every theorem stated below without qualification is proved in the Lean 4 proof assistant [4] against a pinned revision of Mathlib [5]. For each, the compiled theorem’s axiom footprint, reported by `#print axioms`, is exactly `{propext, Classical.choice, Quot.sound}` — the standard axioms of Mathlib itself — and there is no use of `sorry`. Closed forms involving the irrational magnitudes $|k|$ are additionally checked by an independent exact-integer computation (Remark 4.3). The development is public (Appendix).

2. SETUP

Fix $M \geq 1$ and let $\Lambda_M := \{k \in \mathbb{Z}^3 : 0 < |k|^2 \leq M^2\}$, closed under $k \mapsto -k$. A *state* is $u = (u_k)_{k \in \mathbb{Z}^3}$, $u_k \in \mathbb{C}^3$, with $u_k = 0$ for $k \notin \Lambda_M$, $u_0 = 0$, and

$$(\text{divergence-free}) \quad k \cdot u_k = 0, \quad (\text{real}) \quad u_{-k} = \overline{u_k}, \quad \text{for all } k.$$

Here $a \cdot b = \sum_i a_i b_i$ is the *bilinear* dot product (no conjugation) and $\langle a, b \rangle = \sum_i \overline{a_i} b_i$ the Hermitian pairing. Let $\mathbb{P}(k)$ be the orthogonal projector onto $k^\perp$. The truncated Navier–Stokes nonlinearity is

$$(1) \quad B(u, v)_k = \mathbb{P}(k) \left[-i \sum_{\substack{p+q=k \\ p, q \in \Lambda_M}} (q \cdot u_p) v_q \right], \quad k \in \Lambda_M,$$

and the truncated viscous system is $\dot{u}_k = -\nu |k|^2 u_k + B(u, u)_k$, $\nu \geq 0$. Energy and enstrophy are $E(u) = \sum_k |u_k|^2$ and $\Omega(u) = \sum_k |k|^2 |u_k|^2$.

Triad sums. Let $T_M := \{(p, q) \in \Lambda_M \times \Lambda_M : -(p + q) \in \Lambda_M\}$, and for $(p, q) \in T_M$ write $r := -(p + q)$, so $p + q + r = 0$. All “sums over triads” below are sums over the ordered pairs $(p, q) \in T_M$; each unordered triad is therefore counted once per ordered choice of its first two members. The swap $\sigma(p, q) := (p, r)$ is an involution of T_M that fixes p and exchanges q and r .

3. THE WEIGHTED TRIAD IDENTITY, AND ITS TWO INSTANCES

Theorem 3.1 (Weighted triad identity). *Let K be a field in which $2 \neq 0$, let $k_p, k_q, k_r, u_p, u_q, u_r \in K^3$ with $k_p + k_q + k_r = 0$ and $k_p \cdot u_p = 0$, and let $w_q, w_r \in K$. Then*

$$(2) \quad w_r (k_q \cdot u_p)(u_q \cdot u_r) + w_q (k_r \cdot u_p)(u_r \cdot u_q) = (w_r - w_q) (k_q \cdot u_p)(u_q \cdot u_r).$$

Consequently, for a state u on Λ_M (divergence-free suffices; reality is not used) and any weight $w : \Lambda_M \rightarrow \mathbb{C}$,

$$(3) \quad 2 \sum_{(p, q) \in T_M} w(r) (q \cdot u_p)(u_q \cdot u_r) = \sum_{(p, q) \in T_M} (w(r) - w(q)) (q \cdot u_p)(u_q \cdot u_r).$$

Proof. From $k_p + k_q + k_r = 0$ and $k_p \cdot u_p = 0$, $k_r \cdot u_p = -(k_q \cdot u_p)$; the dot product is symmetric, so $u_r \cdot u_q = u_q \cdot u_r$. Substituting gives (2). For (3), the involution σ preserves T_M , so the left side is the sum over T_M of the summand plus its σ -image, which is (2) term by term with $k_p = p$, $k_q = q$, $k_r = r$ (divergence-freeness at p is exactly $p \cdot u_p = 0$). $\square$

Divergence-freeness at p is used exactly once and is necessary: a non-solenoidal field breaks the identity (this is one of the formal development’s negative controls).

Corollary 3.2 (Exact energy conservation). *For every real, divergence-free state u on Λ_M , and every M ,*

$$(4) \quad \operatorname{Re} \sum_{k \in \Lambda_M} \langle u_k, B(u, u)_k \rangle = 0.$$

Proof. Since $u_k \perp k$, the projector drops out of the outer pairing, and reality $\overline{u_k} = u_{-k}$ turns the Hermitian pairing into the bilinear one: $\sum_k \langle u_k, B_k \rangle = -i \sum_{(p,q) \in T_M} (q \cdot u_p)(u_q \cdot u_r)$ with $r = -(p+q) = -k$. This is (3) with $w \equiv 1$, whose right side vanishes; as $2 \neq 0$ in $\mathbb{C}$, the sum is 0. $\square$

Corollary 3.3 (Exact enstrophy production). *For every real, divergence-free state u on Λ_M ,*

$$(5) \quad 2 \sum_{k \in \Lambda_M} |k|^2 \langle u_k, B(u, u)_k \rangle = -i \sum_{(p,q) \in T_M} (|r|^2 - |q|^2) (q \cdot u_p)(u_q \cdot u_r), \quad r = -(p+q).$$

The physical production rate $\frac{1}{2} \frac{d}{dt} \Omega|_{\text{nonlinear}}$ is the real part of the left side.

Proof. As in Corollary 3.2, with the weight $w(r) = |r|^2 = |k|^2$ attached to the third member; apply (3). $\square$

Equations (4) and (5) are the same computation with and without the weight. The whole difference between conservation and production is the factor $|r|^2 - |q|^2$: a triad transfers enstrophy in proportion to the disparity in wavenumber of its two swapped members, and none if they share a sphere. The formal development also checks that the right side of (5) is nonzero on an explicit Galerkin state, so the identity is not vacuous.

Corollary 3.4 (Balance laws). *Let $F_k = -\nu |k|^2 u_k + B(u, u)_k$ with $\nu \geq 0$. For every real, divergence-free state,*

$$(6) \quad \operatorname{Re} \sum_k \langle u_k, F_k \rangle = -\nu \sum_k |k|^2 |u_k|^2 \leq 0,$$

$$(7) \quad \operatorname{Re} \sum_k |k|^2 \langle u_k, F_k \rangle = -\nu \sum_k |k|^4 |u_k|^2 + \operatorname{Re} \sum_k |k|^2 \langle u_k, B(u, u)_k \rangle.$$

Equation (6) says the truncated energy is non-increasing; (7) exhibits the production term of (5) as the sole obstruction to the same statement for the enstrophy. Nothing here bounds that term against the dissipation $\nu \sum_k |k|^4 |u_k|^2$, at fixed M or uniformly; see Section 7.

4. THE HELICAL DECOMPOSITION AND ITS INTERACTION COEFFICIENT

Following Waleffe [2], each transverse plane $k^\perp$ is spanned by two helical vectors $h^\pm(k)$ with $i k \times h^s(k) = s |k| h^s(k)$, $s \in \{+1, -1\}$. We fix them in a way suited to exact arithmetic.

Definition 4.1 (Helical vectors in a frame). *For an integer vector $N \perp k$, $N \neq 0$, set*

$$h_N^s(k) := N \times k + i s |k| N \in \mathbb{C}^3, \quad s \in \{\pm 1\}.$$

For $N \perp k$ these satisfy $h_N^s(k) \cdot h_N^s(k) = 0$, $\overline{h_N^s(k)} = h_N^{-s}(k)$, $k \cdot h_N^s(k) = 0$ and $i k \times h_N^s(k) = s |k| h_N^s(k)$ (all machine-checked). They are Waleffe's $h^s(k)$ multiplied by $|N||k|$: keeping them unnormalised removes every square root and every division from the proofs (in a proof assistant, $x/0 = 0$ silently proves a different theorem). For a triad $p+q=k$ we use the *common* frame vector $N := p \times q$, which is an integer vector orthogonal to p , q , and k , and vanishes exactly on collinear triads.

Expanding $u_k = \sum_s a_k^s h^s(k)$, projecting $B(u, u)_k$ onto $h^s(k)$, and antisymmetrising the two orderings of the triad (the same swap as in Theorem 3.1) produces, for each triad and each chirality triple (s_k, s_p, s_q) , the coefficient

$$(8) \quad C = -\frac{1}{4} (s_p |p| - s_q |q|) g, \quad g = (h_N^{s_p}(p) \times h_N^{s_q}(q)) \cdot \overline{h_N^{s_k}(k)},$$

so that the helical amplitude equations read $\dot{a}_k^{s_k} = \sum_{p+q=k} \sum_{s_p, s_q} C a_p^{s_p} a_q^{s_q} + \dots$ up to the normalisation of the projection. In the frame and scaling of Definition 4.1, g is $|N|^3 |p| |q| |k|$ times Waleffe's geometric factor. A nonzero scaling does not affect any statement about the vanishing of C .

Theorem 4.2 (Closed form in the triad frame). *For $p + q = k$, $N = p \times q$, and any real s_k, s_p, s_q ,*

$$(9) \quad g = -i (N \cdot N)^2 (s_p |p| + s_q |q| + s_k |k|), \quad C = \frac{i}{4} (N \cdot N)^2 (s_p |p| + s_q |q| + s_k |k|) (s_p |p| - s_q |q|).$$

Proof sketch. Expand (8) with Definition 4.1. The terms quadratic in i are each a triple product $(N \times x) \cdot N = 0$, so i survives only linearly and no use of $i^2 = -1$ is needed. The chirality-free term is $(N \times p) \times (N \times q) = (N \cdot (p \times q)) N$, parallel to N , hence annihilated by pairing against $N \times k \perp N$; the remaining terms collapse via $N \times (N \times x) = (N \cdot x) N - |N|^2 x$ with $N \perp p, q, k$. What is left is a degree-9 polynomial identity in six integer variables and the three formal magnitudes, which the proof assistant closes by `ring`. $\square$

Remark 4.3 (Independent verification, and the error it caught). The prefactor and the phase of C depend on the frame and scaling; the formal development proves that in the common frame N the coefficient is purely imaginary with no further hypothesis, and that a global choice of frame (one $N(k)$ per wavevector) transforms differently under $k \rightarrow -k$. The zero locus does not depend on either. Our first derivation of (9) carried $-s_k |k|$; a triad-by-triad comparison against a brute-force construction of h^s and of g from (8) agreed in all eight chirality classes but with two magnitudes permuted — the signature of a sign inside a factor. The corrected form agrees with brute force to 2.1×10^{-16} in floating point and, because both sides are polynomials in $|p|, |q|, |k|$ with integer coefficients, is verified as an *exact integer* polynomial identity over 5 256 triad-by-chirality instances, with an orientation flip $p \times q \rightarrow q \times p$ as the sharpest of three negative controls shown to break it. A formal proof certifies the proof, not the transcription of the statement; the exact-integer check certifies the statement.

Theorem 4.4 (Resonance is collinearity). *Let $p, q \in \mathbb{R}^3 \setminus \{0\}$ with $k = p + q \neq 0$. Then $s_k |k| + s_p |p| + s_q |q| = 0$ for some $(s_k, s_p, s_q) \in \{\pm 1\}^3$ if and only if $p \times q = 0$.*

Proof. The magnitudes are positive, so the signs are not all equal; up to a global sign, exactly one is negative, and the equation is one of $|k| = |p| + |q|$, $|p| = |k| + |q|$, $|q| = |k| + |p|$ — the equality cases of the triangle inequality for $k = p + q$, $p = k + (-q)$, $q = k + (-p)$, each of which holds iff the two summands are parallel and like-directed, i.e. iff $p \times q = 0$. Conversely, if $p \times q = 0$ then p and q are parallel and one of the three equalities holds with the signs chosen accordingly. $\square$

The theorem uses nothing about the lattice; it is stated over $\mathbb{R}^3$ and applies to $\mathbb{Z}^3$ by restriction. Its content for the decomposition is negative: the resonance condition selects no triads beyond those already degenerate (no transverse plane), so it is not an independent locus. On the lattice it was first established by exhaustive exact-integer enumeration over 558 090 triads before being proved.

5. THE VANISHING LOCUS, EXACTLY

Theorem 5.1 (Vanishing locus of the helical coefficient). *For $p + q = k$ in $\mathbb{Z}^3$, $N = p \times q$, and real s_k, s_p, s_q : $C = 0$ if and only if*

$$p \times q = 0 \quad \text{or} \quad s_p |p| + s_q |q| + s_k |k| = 0 \quad \text{or} \quad s_p |p| = s_q |q|.$$

If moreover $p, q, k \neq 0$ and $(s_k, s_p, s_q) \in \{\pm 1\}^3$, the middle alternative implies the first, so

$$C = 0 \iff \text{the triad is collinear, or } s_p |p| = s_q |q|.$$

Proof. By (9), C is $\frac{i}{4} (N \cdot N)^2$ times two real factors; $\mathbb{C}$ has no zero divisors, so $C = 0$ iff $N \cdot N = 0$ (i.e. $N = 0$, as $N \in \mathbb{Z}^3$) or one factor vanishes. The second claim is Theorem 4.4. $\square$

The balance locus is genuinely present and non-degenerate: $p = (1, 0, 0)$, $q = (0, 1, 0)$ have $|p| = |q|$ and $p \times q \neq 0$, and more generally every pair of modes on a common lattice sphere with equal chirality lies on it. What Theorem 5.1 adds to the classical sufficient conditions is the “only if”: no further vanishing configuration exists. The three sufficient conditions had first been proved separately, by degenerating the frame; the closed form re-derives all three algebraically without mentioning a degenerate frame, and the formal development checks on every build that the two routes agree — a check the erroneous first derivation of Remark 4.3 would have failed.

6. THE RESONANT-TRIAD INCIDENCE STRUCTURE ON A PERIODIC LATTICE

We now leave the coefficient and consider only which triads exist. On a periodic lattice, every ordered pair of nonzero modes u, v with $u + v \neq 0$ closes a triad, and the incidence structure is exactly solvable.

Theorem 6.1 (Exact spectrum). *Let G be a finite abelian group with no element of order 2, $\Lambda = G \setminus \{0\}$, $n := |\Lambda|$. For distinct $u, v \in \Lambda$ let $A(u, v)$ be the number of pairs (triad, ordered pair of slots) in which u and v occupy two slots of a triad $a + b = c$ of elements of Λ (so that the coincidence $v = 2u$ is counted, not excluded), and $A(u, u) := 0$. Then*

$$A(u, v) = \begin{cases} 4, & u + v = 0, \\ 6, & \text{otherwise,} \end{cases} \quad \text{i.e.} \quad A = 6(J - I) - 2P,$$

with J the all-ones matrix and P the involution $v \mapsto -v$ (fixed-point free by the 2-torsion hypothesis). Hence every row sum is $6n - 8$; a vector f with $\sum_{\Lambda} f = 0$ and $f(-v) = f(v)$ satisfies $Af = -8f$, and one with $f(-v) = -f(v)$ satisfies $Af = -4f$; the spectrum of the degree-normalised matrix $A/(6n - 8)$ is $\{1, -4/(6n - 8), -8/(6n - 8)\}$, and its normalised gap $1 - \lambda_2$ tends to 1 as $n \rightarrow \infty$.

Proof. The three slot-pair types contribute 2, 2, and $2 - 2[u + v = 0]$ respectively (the last because the triads $(u, v, u + v)$ and $(v, u, u + v)$ require $u + v \neq 0$), giving the stated weight; this is a finite case analysis that the proof assistant discharges. Row sums: $6(n - 1)$ from $6(J - I)$, minus 2 from the single entry of P , which lies off the diagonal since $-u \neq u$. For the eigen-relations, $Jf = 0$ when $\sum f = 0$, so $Af = -6f - 2Pf$, and $Pf = \pm f$. Normalising by the degree gives the spectrum; $\lambda_2 = -4/(6n - 8) \rightarrow 0$. $\square$

The hypothesis is necessary: with an element of order 2, $u = -u$ and the P -term would fall on the excluded diagonal, changing the row sum. The intended instance $(\mathbb{Z}/3)^3$ satisfies it; a 2-torsion counterexample is recorded in the formal development.

Open Problem 6.2 (The spherical truncation). Define the same incidence weight on the spherical truncation $\Lambda_M \subset \mathbb{Z}^3$, where a pair u, v closes a triad only if the third member also lies in the ball. The normalised gap $1 - \lambda_2$ of the resulting matrix, computed exactly, is

M	2	3	4	5	6	7
$ \Lambda_M $	32	122	256	514	924	1418
$1 - \lambda_2$	0.834985	0.833575	0.833419	0.833392	0.833359	0.833348
$(1 - \lambda_2) - 5/6$	1.7×10^{-3}	2.4×10^{-4}	8.6×10^{-5}	5.9×10^{-5}	2.5×10^{-5}	1.5×10^{-5}

(six points, strictly decreasing; at $M = 7$ there are 939 930 triads). We conjecture $1 - \lambda_2 \rightarrow 5/6$: the deviation from Theorem 6.1’s limit 1 would then be a pure boundary effect of the sphere cutoff. Heuristically, in the continuum limit the matrix becomes the kernel $2 \cdot \mathbf{1}_B(x + y) + 4 \cdot \mathbf{1}_B(x - y)$ on the unit ball B , and the conjecture is that its second normalised eigenvalue is $1/6$; we have not proved this and would welcome a direct attack.

7. OPEN PROBLEMS AND DIRECTIONS

- (1) **The spherical gap (Open Problem 6.2).** An exactly solved periodic analogue sits next to a cleanly posed, numerically stable conjecture; the boundary correction may itself be exactly computable.
- (2) **The weighted identity as a tool.** Theorem 3.1 uses only that the observable is a triad sum with one distinguished divergence-free index and a weight on the other two. The same computation should isolate the exact production mechanism, in the form of (5), for other triad-structured quadratic systems — shell models, surface quasi-geostrophic flow, other Galerkin truncations — once the right weight is identified.
- (3) **Other bases.** Theorem 5.1 rests only on the factorisation (9) and the absence of zero divisors; the argument should transfer to other chirality-adapted bases and to $\mathbb{R}^3$, where Theorem 4.4 already holds.
- (4) **Bounds uniform in M .** Everything above is exact at fixed M and gives no estimate. Whether the production term of (7) can be dominated by the dissipation uniformly in M is the question that connects this system to global regularity for the untruncated equations, and it is open. One route in which exact identities of the present kind are the natural input is the self-consistent a-priori-bounds method of rigorous numerics for dissipative PDEs [3]: assume an explicit tail envelope, show dissipation dominates the forcing of every tail mode, conclude invariance. We have not carried this out and make no claim about it.

APPENDIX: REPRODUCIBILITY

The Lean 4 development, the exact-arithmetic cross-checks (Python, exact integers and rationals, each negative control implemented and shown to fail), and the enumerations behind Open Problem 6.2 are in the public repository cited below.¹ The correspondence between statements and formal names is:

- Theorem 3.1: `AbstractAlgebraicConservation.lean`, `weighted_triad_pairing` and `weighted_triad_sum`.
- Corollaries 3.2–3.4: `FourierDynamicsZ3.lean`, `energy_conservation`, `enstrophy_production_identity`, `energyRateZ3_eq`, `enstrophyRateZ3_eq`.
- Theorems 4.2–5.1: `HelicalBasis.lean`, `g0f_closed_form`, `c0f_closed_form`, `resonance_implies_collinear_full`, `c0f_eq_zero_iff`.
- Theorem 6.1: `TriadTorus.lean`, `A_eq`, `row_sum`, `eigen_even`, `eigen_odd`.

Axiom footprints are reproduced by `#print axioms` on each name.

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- [4] L. de Moura, S. Ullrich, The Lean 4 theorem prover and programming language, *CADE-28* (2021).
- [5] The mathlib Community, The Lean mathematical library, *CPP 2020*, DOI 10.1145/3372885.3373824.

¹<https://github.com/xaviercallens/SocrateAI-Scientific-MechanicaFluidorum>, commit 1aa60e0 or later.